PAVAN DHARMA ADAPA

Research Data Analyst · Biomedical Informatics

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PROFESSIONAL SUMMARY

Research data analyst with an M.S. in Computer Science, a year as a graduate research assistant on a public-health study, and co-authorship on a review manuscript under revision. Turns fragile research code into tested, documented, reproducible tools, and is careful about provenance and negative results. Work-authorized through June 2027 on F-1 OPT and eligible for the 24-month STEM extension: no sponsorship required until 2029. Available immediately.

TECHNICAL SKILLS

Research Data Practice: Data management and procurement, survey-weighted estimation, validation against independent implementations, provenance and documentation, manuscript preparation

Analysis & Programming: Python (pandas, NumPy, SciPy, scikit-learn), R, SQL, Advanced Excel, statistical modeling

Scientific Computing: Oxford Nanopore 16S rRNA sequencing, NCBI reference databases, Snakemake workflows, Git and continuous integration, Linux, macOS, WSL2

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University*, Starkville, MS — Jul 2026 – Present

- Rebuilt the lab's DNA sequencing pipeline into a documented, tested tool that installs in one command — **cutting setup from several days to under an hour** — across 32 merged pull requests covering continuous integration, packaging, and cross-platform support.
- Validated pipeline output against an independently built implementation, **agreeing to within 0.01 percentage points** on every species and genus — the verification that made the lab's results defensible to reviewers.
- Processed and quality-controlled **2M+ sequencing reads across 19 samples**, and automated construction of the NCBI reference database into a single command.

Graduate Teaching Assistant | *Mississippi State University*, Starkville, MS — Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**, and reported completion and grade metrics that let instructors reach at-risk students earlier.

Graduate Research Assistant | *Mississippi State University*, Starkville, MS — Aug 2024 – Aug 2025

- Managed and analyzed a **900,000+ record** public-health surveillance dataset, and prepared the written and oral reports presented to faculty across two departments.
- Co-author (third of five) on a review manuscript under revision after peer review; contributed the data preparation, statistical modeling and results reporting, and drafted technical sections.
- Built the automated collection and cleaning workflow that standardized multi-source research data, **cutting per-dataset processing from hours to under 10 minutes**.

HEALTH DATA PROJECTS

Emergency Department Throughput — *CDC/NCHS national survey microdata*

- Analyzed **16,025 real ED visits** survey-weighted to 155,397,747 national visits; the pipeline refuses to run unless it reproduces CDC's published total, so the figures cannot drift undetected.
- Showed between-hospital variation in door-to-provider time is **3.56x larger** than between-hour variation, and reported the expected peak-hour effect as a null result rather than dropping it.

Clinical Quality Measures Engine — *HEDIS / CMS eCQM · synthetic non-PHI EHR (Synthea)*

- Computed HEDIS and CMS eCQM measures over a **13,810-patient EHR** (820,993 encounters, 10.8M observations), holding each measure as a versioned specification with its SNOMED CT and LOINC value sets separate from the engine — so an annual steward update is a reviewable diff a clinical analyst signs off, not a code change.
- Decomposed a diabetes control measure to show **203 of 204 failures were patients never tested**, reframing it as a testing-outreach problem rather than a glycemic-control one.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS — May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India — May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.